5 One-variable descriptive statistics

Sheet 25 Thu 17 Dec. 2026 *Mean and linearity of the mean*

1. Warm-up · Quick [STA] Mean of 12, 15, 9, 14?

   Ans. $\frac{50}{4} = 12.5$.

2. Core · Method [STA] The series 4, 6, 8, 10, 12 has mean 8. What is the mean if 3 is added to each value? If each value is multiplied by 2?

   Ans. Linearity: $8+3=11$; $2\times 8=16$.

   No need to recompute the sum.

3. Core + · True/false [STA] True or false? Justify. The mean of two numbers $a < b$ is always strictly between a and b.

   Ans. True. $a < \frac{a+b}{2} < b$ because $b-a>0$.

4. Review · Reading [LEC] In a pie chart of 80 people, one sector represents 25%. How many people does it represent? What is its angle?

   Ans. 20 people; $90°$.

5. Thread · Quick [DIV] Calculate $\gcd(84,126)$ and deduce the simplest form of $\frac{84}{126}$.

   Ans. $\gcd = 42$; $\frac{84}{126} = \frac{2}{3}$.

Sheet 26 Mon 4 Jan. 2027 *Median and quartiles*

1. Warm-up · Quick [STA] Order 7, 2, 9, 4, 5, 11, 3 then give the median.

   Ans. 2 ; 3 ; 4 ; 5 ; 7 ; 9 ; 11; median 5 (rank 4).

2. Core · Method [STA] Series 3 ; 5 ; 6 ; 8 ; 9 ; 12 ; 14 ; 15. Give Q_1, the median, Q_3 and the interquartile range.

   Ans. $n=8$: Q_1 has rank $\lceil 8/4\rceil = 2$: 5; median $\frac{8+9}{2} = 8.5$; Q_3 has rank 6: 12; $Q_3 - Q_1 = 7$.

   Convention: the smallest value such that at least 25% (resp. 75%) of the data are less than or equal to it.

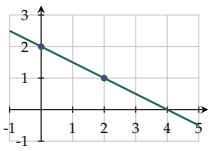
3. Core + · True/false [STA] True or false? Justify. At least half of the values of a series are greater than or equal to its median.

   Ans. True. By definition of the median (at least 50% of the values on each side, endpoints included).

4. Review · Quick [CMP] Put 0.6, $\frac{5}{8}$ and 61% in increasing order.

   Ans. $0.6 < 61\% = 0.61 < \frac{5}{8} = 0.625$.

5. Thread · Graph → formula [REG] The line below represents an affine function f. Determine $f(x)$.



   Ans. It passes through $(0,2)$ and $(2,1)$: slope $\frac{1-2}{2-0} = -\frac{1}{2}$, $f(x) = -\frac{1}{2}x + 2$.

Sheet 27 Tue 5 Jan. 2027 *Standard deviation; effect of adding or removing a value*

1. Warm-up · Quick [CAL] Calculate $(3-5)^2+(7-5)^2$.

   Ans. $4+4=8$.

2. Core · Method [STA] Series 2 ; 4 ; 6: mean, variance and standard deviation (approximate value to 10^{-2}).

   Ans. Mean 4; variance $\dfrac{4+0+4}{3}=\dfrac{8}{3}$; standard deviation $\sqrt{\dfrac{8}{3}}\approx 1.63$.

3. Core + · True/false [STA] True or false? Justify. Adding 5 to each value of a series does not change its standard deviation.

   Ans. True. All the values and the mean are shifted by 5: the deviations from the mean are unchanged.

4. Review · Construction [CON] Let ABC be a triangle, I the midpoint of $[AB]$ and D the reflection of C in I. What type of quadrilateral is $ACBD$?

   Ans. A parallelogram: its diagonals $[AB]$ and $[CD]$ have the same midpoint I.

5. Thread · Quick [GEO] $(MN)\parallel(BC)$, $M\in[AB]$, $N\in[AC]$, $AM=3$, $AB=5$, $AC=10$. Calculate AN.

   Ans. Thales: $\dfrac{AN}{AC}=\dfrac{AM}{AB}$, so $AN=10\times\dfrac{3}{5}=6$.

Sheet 28 Thu 7 Jan. 2027 *Comparing series*

1. Warm-up · Quick [STA] Range of 3, 8, 5, 12?

   Ans. $12-3=9$.

2. Core · Interpret [STA] Two series A and B have the same mean 12; $\sigma_A=1.5$ and $\sigma_B=4$. Which one is more consistent?

   Ans. A: the values are closer together around the mean (smaller σ).

3. Core + · Interpret [STA] A: median 11, interquartile range 2. B: median 13, interquartile range 6. Compare the two series.

   Ans. B is centred higher (median $13>11$) but more spread out (interquartile range $6>2$).

4. Review · Quick [PRB] Two fair coins are tossed. What is the probability of getting two heads?

   Ans. 4 equally likely outcomes: $\dfrac{1}{4}$.

5. Thread · Method [ALG] Programme: choose a number; square it; add 1. Result for -3? Can 0 be obtained? Justify.

   Ans. For -3: $(-3)^2+1=10$. No: $x^2\geqslant 0$ so $x^2+1\geqslant 1>0$.

Sheet 29 Mon 11 Jan. 2027 *Grouped data: histogram and cumulative frequencies*

1. Warm-up · Quick [STA] Give the width and the midpoint of the class $[20,35)$.

   Ans. Width 15; midpoint $\dfrac{20+35}{2}=27.5$.

2. Core · Table [LEC] Classes $[0,10)$, $[10,20)$, $[20,30)$ with frequencies (counts) 4, 6, 10. Give the relative frequencies, then the cumulative increasing relative frequencies.

   Ans. Total 20; relative frequencies 20%, 30%, 50%; cumulative 20%, 50%, 100%.

3. Core + · Quick [LEC] In a histogram with classes of equal width, two rectangles have heights 3 cm and 6 cm. What is the ratio of their frequencies (counts)?

   Ans. The heights are proportional to the counts: 1 to 2.

4. Review · Quick [NUM] Write 0.00032 in scientific notation.

   Ans. 3.2×10^{-4}.

5. Thread · Quick [EQ] Solve in $\mathbb{R}$: $2(x+3)=5x-9$.

Ans. $x = 5$
Pitfall: forgetting to distribute the 2 over the 3.

Sheet 30 Tue 12 Jan. 2027 *Weighted mean, median class; spreadsheet and Python*

1. Warm-up · Quick [CAL] Calculate $\dfrac{2 \times 8 + 3 \times 12}{5}$.

   Ans. $\dfrac{52}{5} = 10.4$.

2. Core · Method [STA] Classes $[0, 10)$, $[10, 20)$, $[20, 30)$ with frequencies (counts) 4 ; 7 ; 9. Estimate the mean using the class midpoints; give the median class.

   Ans. Estimated mean $\dfrac{4 \times 5 + 7 \times 15 + 9 \times 25}{20} = 17.5$; cumulative frequencies 4 ; 11 ; 20: the 10^{e} and 11^{e} values are in $[10, 20)$: median class.

3. Core + · Code reading [ALG] What does this programme calculate and what value does it give?

```
notes = [12, 15, 9, 14]
s = 0
for n in notes:
    s = s + n
print(s / len(notes))
```

   Ans. The mean of the marks: $\dfrac{50}{4} = 12.5$.

4. Review · Quick [PRO] An item costing 45,000 Ar is sold at -30%. What is the sale price?

   Ans. $45{,}000 \times 0.70 = 31{,}500$ Ar.

5. Thread · Prove [LIT] Programme: choose n; add 1; square it; subtract n^2. Show that the result is always an odd number.

   Ans. $(n+1)^2 - n^2 = 2n + 1$: odd for every integer n.